DSA

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PART A PROBLEM SELECTION

APPLICATION: EMPLOYEE ID SEARCH

(A) Problem:

A company must store employee records (ID, name, department), add new employees, and retrieve any record by ID.

(B) Why efficient searching:

Payroll, attendance and login system loop us IDs constantly. A linear scan costs O(n) per lookup, which becomes too slow as the employee count grows.

(C) Why compare these three:

- BST: simple, O(log n) on average, but can degrade to O(n).
- AVL: a BST that balances itself, giving guaranteed O(log n).
- Hashing: O(1) on average, with linear probing to resolve collisions.

Together they show the trade off between simplicity, guaranteed Performance and raw speed.

PART B: Implementation

- BST: insert, search, in order / Preorder / Postorder, height
- AVL: insert with LL, RR, LR and RL rotations, search, height
- Hashing: Key % 61, linears probing (i+1) % 61, collision counter.

PART C: Mini Experiment (random insertion order)

Observation	BST	AVL	Hashing
successful search, all 40 keys (total)	225	184	56
Successful search (avg/ max)	5.62/9	4.6/6	1.4/6
unsuccessful search, 9 absent keys (avg/max)	4.67/7	5.44/6	3.78/7
Insertion, building 40 key (total)	185	160	56
Inserting 4 new key (1111, 1222, 1333, 1555)	6,6,4,7	6,6,6,5	5,4,1,1

PART D: ALGORITHM BATTLE

Parameter	BST	AVL	Hashing
search comparison	Avg 5.62	Avg 4.6	Avg 1.4
Insertion observations	185 comparisons, no rebalancing	160 comparisons 26 rotations	56 probes, no rotations
Height	9 (40 on sorted input)	6 (6 on sorted input)	—
collisions	—	—	14 (load factor 0.60)
overall observations	Fine on random data, collapses on sorted data (avg 20.5 comparisons)	stays balanced on any input	fastest, but unordered and slows as the table fills.

PART E. Final verdict:

(A) Winner: Hashing with linear probing

(B) Sorting Observations:

* It needed the fewer comparisons for successful search (1.40, against 4.60 for AVL and 5.62 for BST).

* It was cheapest to build (56 probes, against 160 & 185)

* It also did best on unsuccessful search (3.78 against 4.67 to 5.44
* Its cost was 16 collisions at a load factor of 0.66 which is acceptable if you reach above 0.7.